Assessment Cover Sheet

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ASSESSMENT DETAILS

Unit title	SOFTWARE ARCHITECTURE AND DESIGN	Tutorial /Lab Group	1	Office use only
Unit code	SWE30003	Due date	2/10/2026	
Name of lecturer/tutor	Prof Ts Dr Lau Bee Theng			
Assignment title	ASSIGNMENT 1 SRS			Faculty or school date stamp

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1.Introduction

The Sarawak Tourism Promotion System (STPS) is a software system for promoting tourism in Sarawak. It gives a tourist a platform to find tourism information. A tourist can search for a destination (park, cultural site, festival), lodging listing (hotel, homestay, resort), vehicle listing (car, van, boat, bus), food establishment (restaurant, cafe, food court), or local information (culture, safety, weather). A tourist can also plan a trip, make a booking, make a payment, and give a review. A management staff member can manage tourism content, listing, booking, and availability.

2. Project Overview

STPS is made for tourists from around the world and a local tourism association. The system puts Sarawak tourism information and service on one platform. A tourist can use the system to find a destination, lodging listing, vehicle listing, food establishment, or local information. A tourist can also plan a trip, make a booking, make a payment, and give a review. A management staff member can manage tourism content, listing, booking, and availability. The system also protects personal information based on Malaysian PDPA requirements.

2.1 Goals

The first goal of STPS is to provide one main digital platform for promoting Sarawak tourism. A tourist can find tourism information and a tourism service in one place.

The second goal is to give the tourism association control over its content and data. A management staff member can create, update, publish, unpublish, and archive a listing.

The third goal is to provide trustworthy and current local information. A tourist can find information about a destination (park, cultural site, festival), culture, safety, weather, or event.

The fourth goal is to protect personal information based on Malaysian PDPA requirements. The system protects information such as a tourist name, contact information, and booking information.

2.2 Assumptions

The project assumes that the required hardware (server, network device), and data repository are already available. The project focuses on developing software.

The system assumes that a tourist has a modern web browser or smartphone with an internet connection.

The project assumes that payment uses an external payment gateway (online payment service). STPS does not store raw payment card information.

The project assumes that the tourism association is responsible for the tourism content. A management staff member can create, update, publish, unpublish, and archive a listing.

The project assumes that English and Bahasa Malaysia are the main languages for the first version. Malaysian PDPA is the main privacy requirement.

2.3 Scope

The scope of STPS covers tourism discovery, trip planning, booking, payment, and review. A tourist can search for local information, view a destination, find a lodging listing, find a vehicle listing, find a food establishment, create a package, make a booking, make a payment, and submit a review.The scope also covers a management function. A management staff member can manage tourism content, a listing, a booking, availability, and a tourist privacy request.

The system can connect to an external service, such as a Payment Gateway Provider (payment service), Notification Service (booking notification), Mapping or Geolocation Service Provider (location service), and Translation or Localization Service (language service).The project does not cover a physical server, network infrastructure, end-user device, or the internal operation of a third-party service. These are outside the software scope of STPS.

3. Problem Domain

3.1 Pain Points

- Unverifiable Tourist Information
    - Tourist information is scattered across informal/unofficial sites and social media, inconsistent and unverifiable.
- No combined Discovery and Booking Channel
    - No single channel that lets a tourist discover and book accommodation, transport, and food together.
- Manual Content Updates
    - Management updates tourism content manually via spreadsheets, printed brochures causing delay and error.
- Language and Currency Barriers
    - Language and currency barriers stop worldwide tourists from using fragmented local-only sites.
- No Structured feedback Channel
    - No feedback channel exists for the association to assess visitor satisfaction or provider quality.
- Ungoverned Personal Data Handling
    - Absence of a governed data-handling process exposes tourist and management personal data to PDPA compliance risk.

3.2 Domain Entities

- Tourist
- Management Staff
- Local Service Provider
- Review
- Booking
- Local Article Info
- Payment
- Services
- Destination
- Vehicle
- Food Establishment
- Accommodation

Actors

- Tourist
    - Primary end user of the platform
    - Can browse as guest or under a registered profile
- Management Staff
    - Content Officer
        - Responsible for curating listings and promotional content
    - Data/System Admin
        - Responsible for account/permission administration and PDPA data-subject request handling.8
- Payment Gateway provider
- Notification Service
- Mapping Service Provider
- Translation/Localization Service

3.3 Tasks

1. Discover Local Tourism Information
2. Reserve Accommodation
3. Reserve Transportation
4. Reserve Food & Dining
5. Book services
6. Submit Review & Feedback
7. Manage Promotional Content & Listing
8. Coordinate Bookings Availability
9. Manage Own Listing Availability
10. Decide on Own-Listing Bookings

4. Functional Requirements and Task Descriptions

Task 1: Discover Local Tourism Information

Task: Discover Local Tourism Information	
Actors	Tourist
Trigger	Tourists want destination, culture, safety, weather, or event information
Precondition	Platform is accessible in guest or registered mode
Postcondition	Tourists have viewed or saved relevant local-information content
Frequency	High
Priority	Medium-High
Critical	This is a core promotional function directly serving the Single Authoritative Promotion Channel and Trustworthy Current Local Information goals.
Work Area	Local information
Solution Agnosticism Check	Searchable web content hub; mobile app content feed; QR-linked kiosk/brochure using the same content repository
Subtasks and System Support	
Subtasks	**System Support**
1. Search information by category such as culture, safety, weather and events. **Problem:** No matching results	Suggest related or alternative categories
2. View a Local Information Article in detail. **Problem:** Article is outdated or unpublished	Automatically hide expired content from tourist-facing views
3. Bookmark an article for later. **Problem:** Guest tries to bookmark	Prompt the Guest to register without losing the current browsing context
Variants	
Variants	**Example of solutions/ System Support**
1a. Tourist searches using location	Show local information related to the Tourist's current location when geolocation is allowed
2a. Tourist has no internet connection	Allow the Tourist to view previously saved local information for offline use
3a. Major festival or event is happening	Show relevant event and festival information first to help the Tourist plan the visit

Task 2: Reserve Accommodation

Task: Reserve Accommodation	
Purpose	Allow tourists to find suitable accommodation, determine its availability for their intended stay, provide the required booking details, and make a reservation.
Trigger / Precondition	The tourist has identified the need for accommodation in Sarawak and has sufficient information about the intended location and travel period to begin a reservation.
Frequency	As required; potentially multiple times during trip planning.
Critical	Accommodation may be unavailable for the selected dates, the requested number of guests may exceed available capacity, or the tourist may need to change the selected accommodation or stay period.
Work Area	Accommodation discovery and reservation area.
Subtasks and System Support	
Subtasks	**System Support**
1. Find and select suitable accommodation	Allow the tourist to specify location, stay period and number of guests, display matching accommodation, provide relevant accommodation information, and show availability for the selected dates and capacity.
2. Provide details and confirm reservation	Allow the tourist to enter and validate required guest and reservation details, record the reservation, and provide confirmation containing the relevant booking information.
Variants	
Variants	**Example of solutions/ System Support**
1a. Accommodation is unavailable or no suitable option is found	Inform the tourist and allow alternative accommodation, dates or requirements to be selected.
2a. Booking information is incomplete or invalid	Identify the missing or invalid information and allow the tourist to correct it before confirmation.
2b. Tourist changes reservation details	Allow the tourist to review and modify the reservation before it is recorded.

Task 3: Reserve Transportation

Purpose	Allow tourists to identify suitable transportation for their travel needs, check availability, provide passenger and journey information, and make a reservation.
Trigger / Precondition	The tourist requires transportation between one or more locations and has sufficient information about the intended journey to begin a reservation.
Frequency	As required; potentially multiple times when planning or undertaking a trip.
Critical	Transportation may be unavailable for the requested time, have insufficient capacity, or does not satisfy the tourist's intended origin, destination or travel requirements.
Work Area	Transportation discovery and reservation area.
Subtasks and System Support	
Subtasks	**System Support**
1. Review transportation information	Provide relevant information about the selected transportation option, such as provider, route or journey details, applicable conditions and available information about cost.
2. Check availability	Allow the tourist to determine whether the selected transportation option is available for the requested journey and passenger requirements.
3. Confirm reservation	Record the transportation reservation and provide the tourist with confirmation containing the relevant journey and booking details.
Variants	
Variants	**Example of solutions/ System Support**
2a. Selected transportation is unavailable	Inform the tourist that the selected option is unavailable for the requested journey and allow alternative transportation or journey details to be considered.
2b. Passenger capacity is insufficient	Inform the tourist that the selected transportation option cannot accommodate the requested number of passengers and allow another option to be selected.
3a. Journey details are changed before confirmation	Allow the tourist to review and modify journey and reservation details before the booking is recorded.

Task 4: Reserve Food & Dining

Task: Reserve Food & Dining	
Actors	Tourist
Trigger	Tourists want to reserve a Food Establishment (restaurant, cafe, or food court) for a date, time, and number of guests.
Precondition	The platform is accessible in guest or registered mode. At least one published Food Establishment listing exists, and the Tourist can state a location or establishment, date and time, and number of guests.
Postcondition	A pending Booking is recorded with a timestamp, within published availability. The Local Service Provider approves or declines it. The Tourist pays only after approval. If the Tourist stops before confirmation, no Booking is recorded.
Frequency	High (peaks during holidays and major events)
Critical	Details may be missing or invalid, no listing may match, or capacity may be insufficient.
Priority	High (supports combined discovery and booking; food is a bookable offering)
Work Area	Food establishment reservation
Solution Agnosticism Check	Web reservation, mobile booking flow, or staff-assisted counter/kiosk, all using the same listings and availability.
Subtasks and System Support	
Subtasks	**System Support**
Specify area or establishment, date and time, and number of guests. **Problem:** Details missing or invalid	The system flags the information for correction before searching or saving.
Find an establishment, review details, and check availability. **Problem:** No listing matches, or not enough remaining capacity	Informs the Tourist and allows changing location, time, or guests; never books beyond remaining capacity, including capacity held by packages with the same establishment
Provide guest/contact details and confirm. **Problem:** Contact details incomplete or invalid	Flags the issue for correction before confirmation; rechecks availability if time or guests change
Variants	
Variants	**Example of solutions/ System Support**
1a. Required details are missing or invalid	Identify the missing or invalid information and allow the tourist to correct it before proceeding
2a. No suitable Food Establishment is found	Inform the tourist that no listing matches and allow the location, time or number of guests to be changed.

Task 5: Book Services

Task	Book Services
Actors	Tourist
Trigger	A Tourist wants to book a Service Package or build their own itinerary from services (Destinations, Accommodation, Vehicles, Food Establishments) as one Booking.
Precondition	The platform is open to guests and registered users. Published Packages and offerings exist, and the Tourist has a travel period.
Postcondition	One Package Booking is saved with a timestamp, along with a component Booking for each Accommodation, Vehicle, and Food Establishment, all within available capacity.
Frequency	Medium (once per trip, with peaks before holidays and major events)
Critical	A needed part of the trip may be missing or unavailable.
Priority	High (the task provides one channel to discover and book accommodation, transport, and food together)
Work Area	Service package and itinerary
Solution Agnosticism Check	Web, mobile, or staff counter/kiosk, all using the same Package and availability data.
Subtasks and System Support	
Subtasks	**System Support**
Choose a Package **Problem:** A required service is missing	Saves the plan and shows matching Packages so the Tourist can add or replace parts
Check availability and price **Problem:** A service is unavailable, full, or changes before confirmation	Checks each component, keeps the rest of the plan, lets the Tourist replace it or change dates, and shows the total in MYR plus an estimate in their currency
Enter contact details and confirm **Problem:** Details are missing or invalid, or the Tourist changes something	Validates details, allows changes, then saves the Booking and shows its status. No card details collected
Variants	
Variants	**Example of solutions/ System Support**
2a. A required part of the trip is missing	Keep the plan, show published Packages that match the travel period, and allow the tourist to add or replace a component before confirming.
4a. A component is unavailable or over capacity	Inform the tourist, keep the rest of the plan, and allow them to replace that component.

Task 6: Submit Review & Feedback

Purpose	Allow tourists to rate and review listings they have used and give feedback, helping others choose and the association monitor quality.
Trigger / Precondition	A signed-in tourist has used a listing or wants to give platform feedback.
Frequency	Medium: after most complete bookings, with peaks after major events.
Critical	Offensive, fake or duplicate reviews, and spikes after major events.
Work Area	Review and feedback area.
Subtasks and System Support	
Subtasks	**System Support**
1. Choose what to review	Show listings the tourists have used, plus a general feedback option.
2. Rate and write the review	Allow a star rating and optional comment.
3. Check eligibility	Confirm a completed booking for the item.
4. Screen the content	Check for offensive language, personal data or spam.
5. Moderate flagged reviews	Let Management Staff approve or reject flagged reviews.
6. Publish the review	Show it as verified, update the average rating and notify the tourist.
7. Use feedback for improvement	Show Management Staff feedback and rating trends.
Variants	
Variants	**Example of solution / System Support**
2a. Tourist has already reviewed the item	Open the existing review for editing instead of adding a duplicate.
3a. Tourist has no completed booking	Explain that reviews need a completed booking; offer general feedback instead.
5a. Moderator rejects the review	Keep it hidden and tell the tourist why, so they can edit and resubmit.

Task 7: Manage Promotional Content & Listing

Purpose	Allow Content Officers to manage listings and promotions so tourists always see accurate, current information.
Trigger / Precondition	New or changed Service Provider details, a planned promotion, or outdated content.
Frequency	Regular: daily to weekly, with peaks before major festivals and holidays.
Critical	Many updates before major festivals, outdated details, and withdrawing listings with active bookings.
Work Area	Content management area (back office).
Subtasks and System Support	
Subtasks	**System Support**
1. Select content to manage	Find content by category, status or Service Provider, or start a new item.
2. Enter or update content details	Enter description, location, prices, dates and images.
3. Provide both language versions	Add English and Bahasa Malaysia text.
4. Validate the content	Check required fields and formats.
5. Preview the content	Show it as tourists will see it.
6. Publish or schedule the content	Publish now or on a set date, log who and when, and notify the Service Provider.
7. Unpublish or archive content	Hide it from tourists, record the reason and keep it for reuse.
8. Keep content current	Flag outdated or expired content.
Variants	
Variants	**Example of solution / System Support**
4a. Details are missing or invalid	Highlight the problems; allow saving as a draft but not publishing.
6a. Promotion is time-limited	Publish and unpublish automatically on the set start and end dates.
7a. Listing has active bookings	Warn and show the affected bookings to resolve first (Task 10).

Task 8: Manage Bookings Availability

Task: Manage Bookings Availability	
Actors	Management Staff
Trigger	A new booking request is received; availability needs adjustment as its Local Service Provider did not act within the defined SLA window (within 48 hours upon booking is created)
Precondition	Staff is authenticated with booking-management permission; Booking(s) exist
Postcondition	A Booking is approved/adjusted/cancelled. Availability calendars are updated accordingly
Frequency	High
Critical	Failures may approve unavailable Bookings, delay refunds, or affect confirmed reservations
Priority	High. Directly protects the Increased Provider Bookings & Revenue and Data-Informed Management Decisions incentives.
Work Area	Booking/ Availability administration module.
Subtasks and System Support	
Subtasks	**System Support**
1.Review pending Package Booking.	
2. Approve/reject a pending Package Booking	
3. Finalize a component Booking that its Local Service Provider did not approve **Problem:** Payment was already made when booking is cancelled	Grants a refund instead of requiring Management Staff to initiate one manually
4. Adjust an availability calendar **Problem:** Staff sets a conflicting availability window overlapping confirmed Bookings	Before saving a newly entered availability calendar change, warns Management Staff if the change would overlap one or more existing Confirmed Bookings.
5. Modify a Booking on the Tourist's behalf. **Problem:** Cancellation occurs after payment	Management Staff modifies a Booking for which the payment has already been made, initiate the refund workflow instead of requiring a separate manual refund request.
Variants	
Variants	**Example of solution / System Support**
Provider decision	If the provider declines or does not decide within the SLA,Management Staff may finalize the non-approval or override it.
Staff cancellation or modification	A cancellation or modification before payment capture does not require a refund workflow
Capacity and calendar change	At zero capacity, an Accommodation, Vehicle, or Food Establishment is shown as "Full" and new requests against that capacity are blocked without staff override

Task 9: Manage Own Listing Availability and Booking Decisions

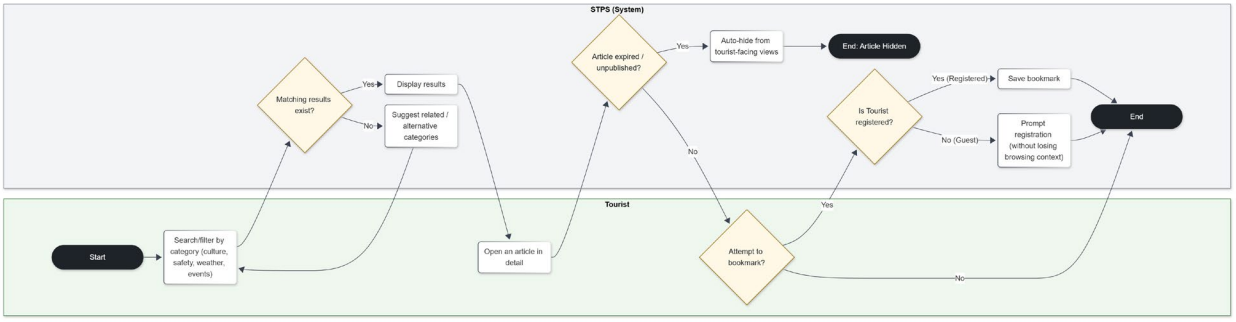
Task: Manage Own Listing Availability	
Actors	Local Service Provider
Trigger	A new Booking request arrives against the provider's own Accommodation, Vehicle, or Food Establishment. The provider needs to set up the day's available capacity. The capacity can be adjusted when required.
Precondition	The provider is authenticated via the scoped self-service login
Postcondition	The day's capacity count is set or updated, reflected immediately in tourist-facing availability
Frequency	High
Critical	Failures may create stale capacity, overbooking, or reductions below confirmed demand
Priority	High. Directly protects the Increased Provider Bookings & Revenue incentive, while a missed or slow decision risks a lost booking for the Tourist.
Work Area	Local Service Provider self-service module
Subtasks and System Support	
Subtasks	**System Support**
1. Set the number of available slots for a given day on an owned listing. **Problem:** Provider sets a slot count below the number of already-Confirmed Bookings for that day	Blocks the change. Then, requires explicit resolution with management rather than silently overbooking or silently cancelling already-confirmed Tourists.
2. Publish updated number of available slots.	
Variants	
Variants	**Example of solution / System Support**
One-day override and recurring pattern	The requirement defines a daily capacity value. Recurrence rules and precedence between a default pattern and a one-day override are not specified and require confirmation before implementation
Capacity increase or safe reduction vs. conflicting reduction	Blocks a change below the number of Confirmed Bookings until the provider explicitly resolves the conflict.

Task 10: Decide on Own-Listing Bookings

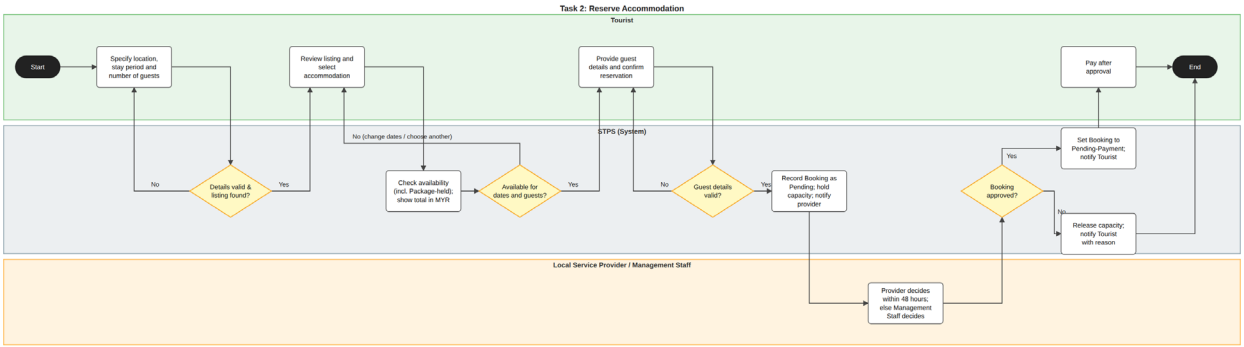
Task: Decide on Own-Listing Bookings	
Actors	Local Service Provider
Trigger	A new Booking request arrives against the provider's own Accommodation, Vehicle, or Food Establishment.
Precondition	The provider is authenticated via the scoped self-service login
Postcondition	The day's capacity count is set or updated, reflected in tourist facing availability
Frequency	High
Critical	Failures may leave Bookings unresolved, delay escalation, or hide rejection reasons.
Priority	High, a missed or slow decision risks a lost booking, irresponsible feedback
Work Area	Local Service Provider self-service module
Subtasks and System Support	
Subtasks	**System Support**
1. Review a pending Booking against an owned listing.	
2. Approve or decline a pending Booking against an owned listing **Problem:** the provider does not act within a defined SLA window	Escalates the pending Booking to Management Staff for override (Subtask 10.3) instead of leaving the Tourist not updated.
3. Receive notification of new pending Bookings and a daily slot-utilization summary. **Problem:** The provider declines a Booking	Requires a decline of reason before completing the action, so both the Management Staff and the Tourist are updated.
Variants	
Variants	**Example of solution / System Support**
Provider approves within the SLA vs. declines within the SLA	Approval confirms the component Booking and updates availability. A decline requires a reason, which is sent to the Tourist and made visible to Management Staff;
Decision within the SLA vs. No decision by the deadline	A timely decision follows the provider's approve/decline path. If no decision arrives within the defined SLA, the system escalates the Booking to Management Staff for override rather than leaving it pending indefinitely

5. Workflow

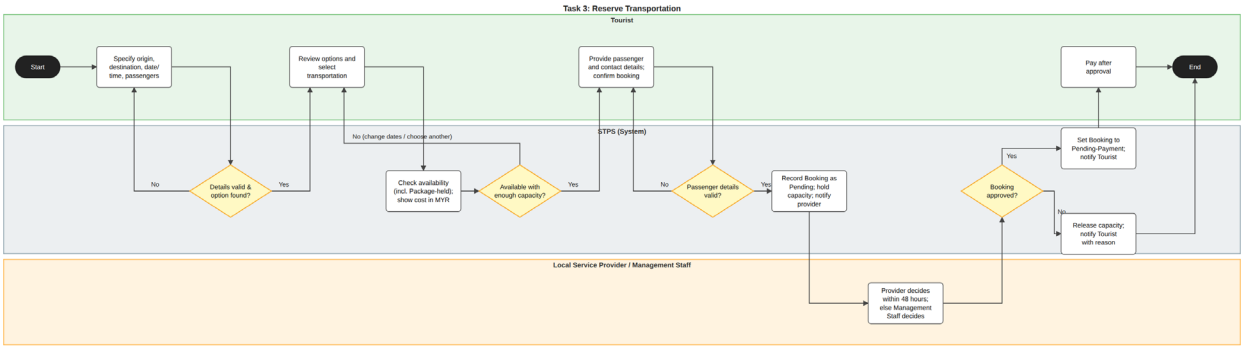
1. Discover Local Tourism Information



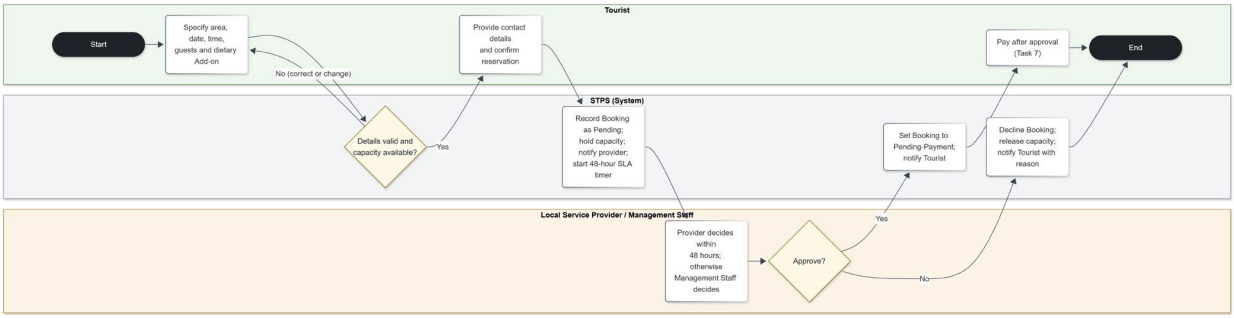
2. Reserve Accommodation



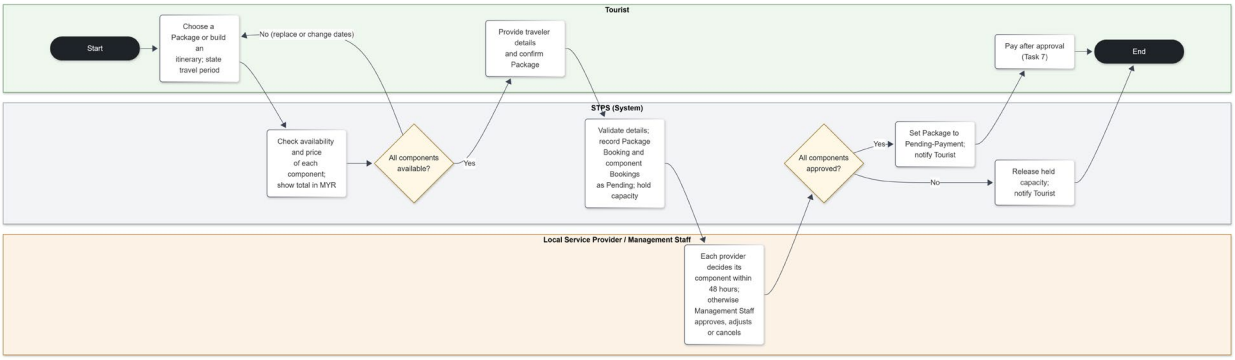
3. Reserve Transportation



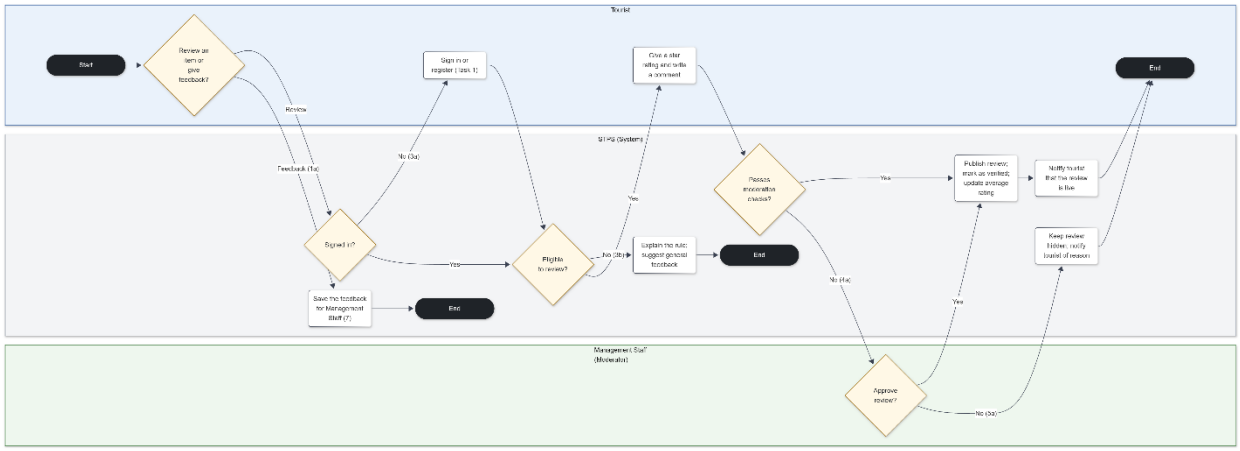
4. Reserve Food & Dining



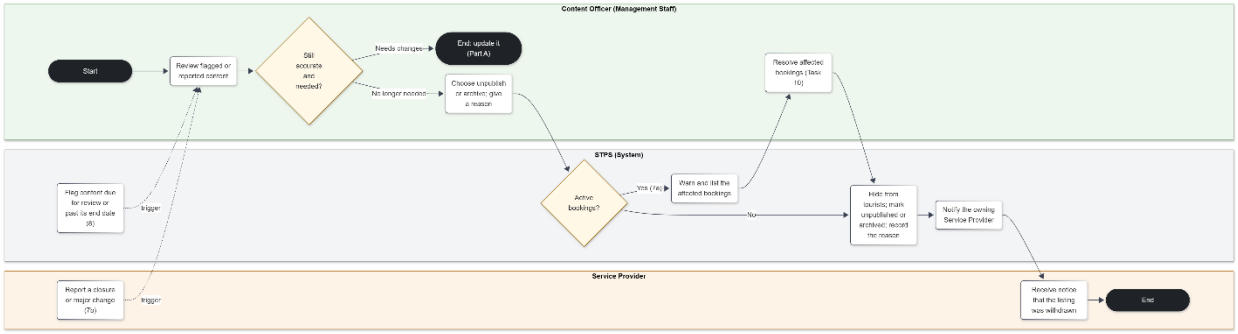
5. Book Services



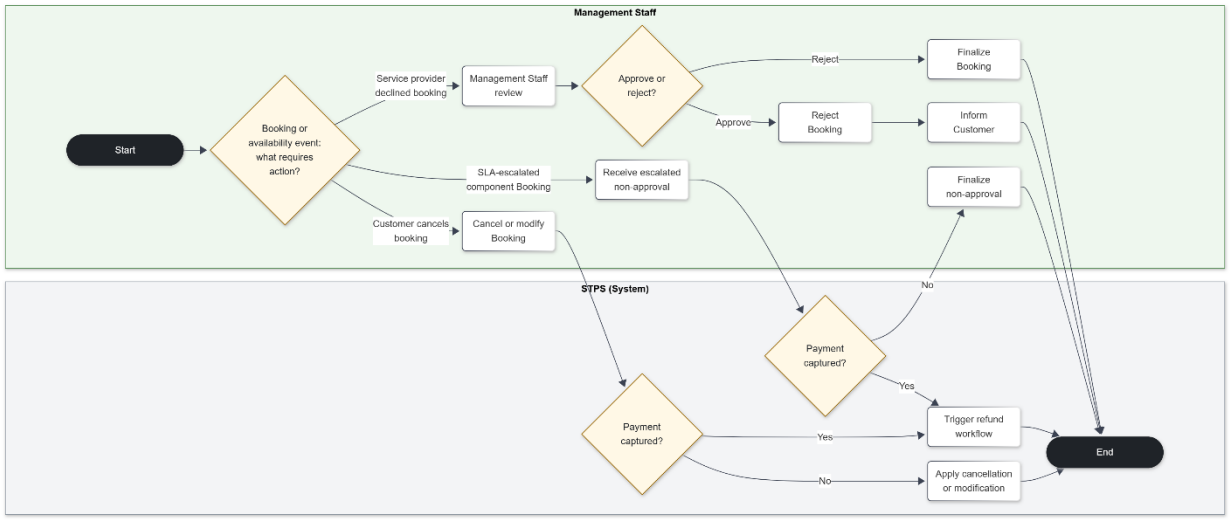
6: Submit Review & Feedback



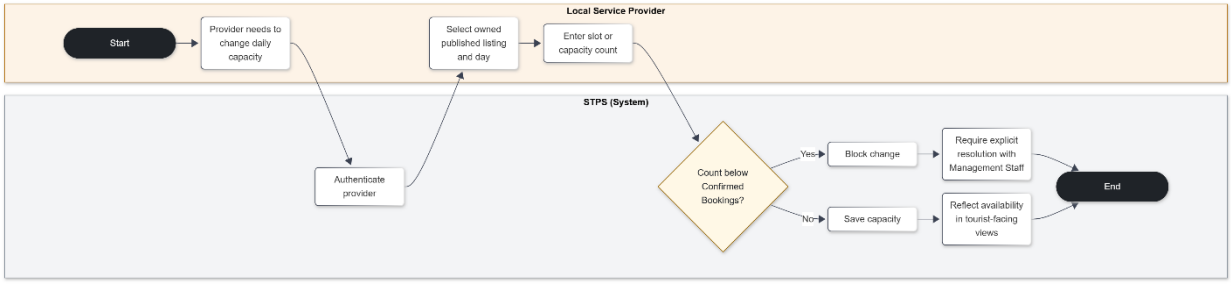
7: Manage Promotional Content & Listing



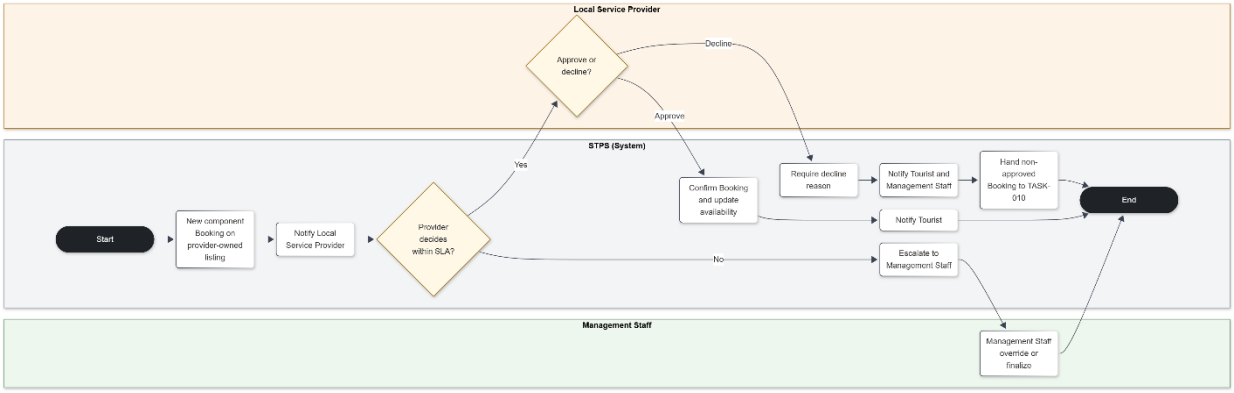
8. Manage Bookings Availability



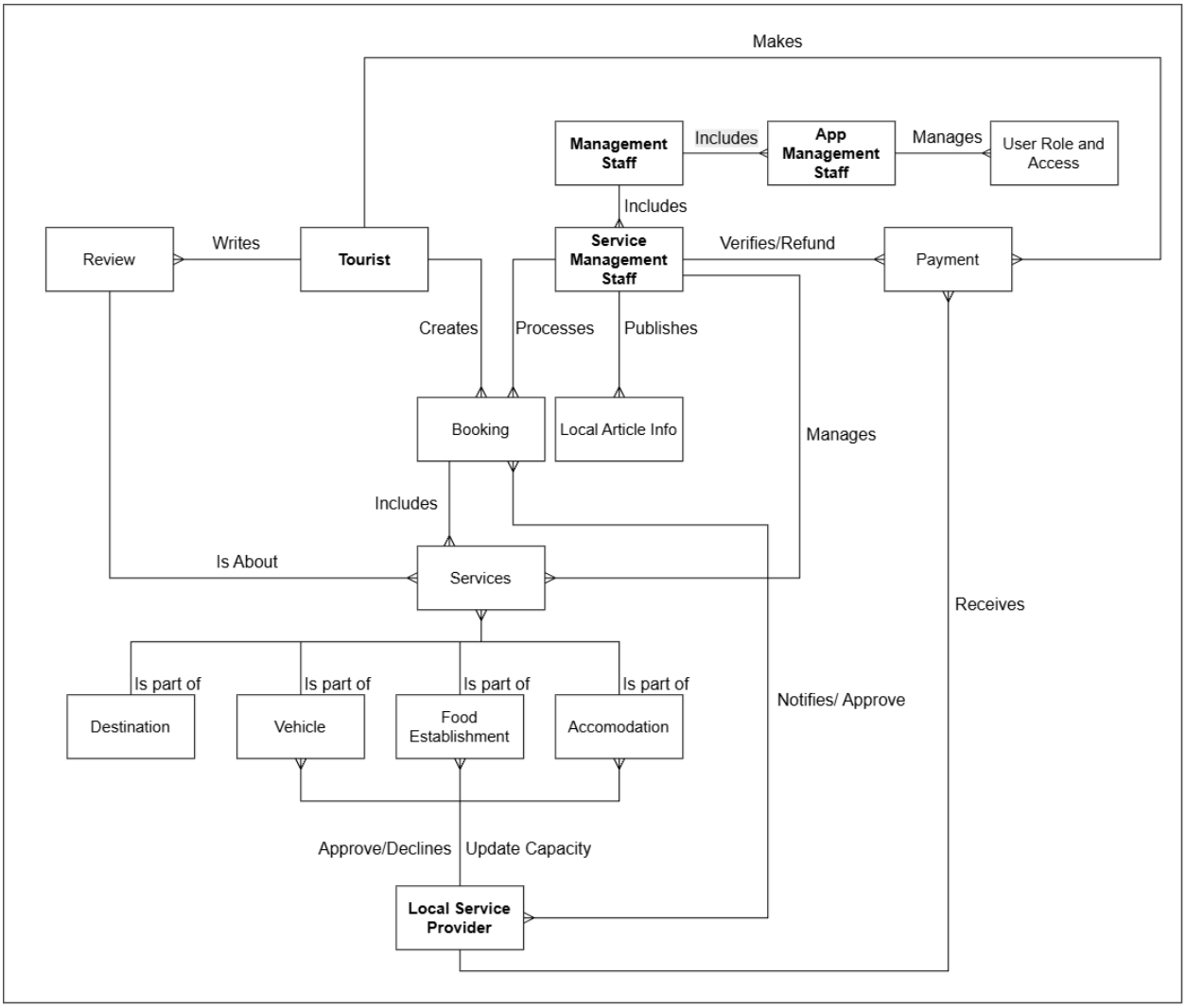
9. Manage Own Listing Availability



10. Decide on Own-Listing Bookings



6. Data Model



Domain Entities

Entity	Description	Key Relationships & Cardinality
Tourist	An individual traveler and system user who interacts with services by planning, reserving, paying, and providing feedback.	• Writes **Review** (One-to-Many) • Creates **Booking** (One-to-Many) • Makes **Payment** (One-to-Many)
Management Staff	A generalized parent entity for administrative users, which is now split into specialized functional roles.	• Includes **App Management Staff** (One-to-Many) • Includes **Service Management Staff** (One-to-Many)

App Management Staff	A specialized administrative role focused on system-level administration and permissions.	• Is included in **Management Staff**
User Role and Access	An entity representing the permissions, security roles, and access levels assigned to users within the system.	• Managed by **App Management Staff** (Many-to-One)
Service Management Staff	A specialized administrative role handling operational workflows, financial verification, and content management.	• Is included in **Management Staff** • Verifies/Refunds **Payment** (One-to-Many) • Processes Booking (One-to-Many) • Publishes **Local Article Info** (One-to-Many) • Manages **Services** (One-to-Many)
Local Service Provider	Third-party business partners (e.g., hoteliers, tour operators) supplying the actual travel and tourism offerings.	• Receives **Payment** (One-to-Many) • Approves/Declines availability for **Vehicle, Food Establishment, Accommodation** (One-to-Many) • Updates capacity for **Vehicle, Food Establishment, Accommodation** (One-to-Many) • Notifies/Approves **Booking** (One-to-Many)
Review	Feedback and ratings authored by travelers regarding their experience with specific offerings.	• Written by **Tourist** (Many-to-Many) • Is about **Services** (Many-to-Many)
Booking	A confirmed reservation transaction made by a tourist for specified dates and services.	• Created by **Tourist** (Many-to-One) • Processed by **Management Staff** (Many-to-One) • Includes **Services** (Many-to-Many)
Local Article Info	Informational content such as travel guides and destination highlights.	• Published by **Management Staff** (Many-to-One)

Payment	A financial transaction record for funds sent by a tourist to cover booking costs.	• Made by **Tourist** (Many-to-One) • Verified/Refunded by **Management Staff** (Many-to-One) • Received by **Local Service Provider** (Many-to-One)
Services	The core generalized entity for any bookable offering on the platform.	• Subject of **Review** (Many-to-Many) • Included in **Booking** (Many-to-Many) • Managed by **Management Staff** (Many-to-One) • Parent entity for **Destination, Vehicle, Food Establishment, Accommodation**
Destination	A sub-type of *Services* representing a specific place of interest or landmark.	• Is part of **Services** • Capacity/Approvals managed by **Local Service Provider**
Vehicle	A sub-type of *Services* representing transport offerings like rentals or tours.	• Is part of **Services** • Capacity/Approvals managed by **Local Service Provider**
Food Establishment	A sub-type of *Services* representing dining facilities and restaurants.	• Is part of **Services** • Capacity/Approvals managed by **Local Service Provider**
Accommodation	A sub-type of *Services* representing lodging properties.	• Is part of **Services** • Capacity/Approvals managed by **Local Service Provider**

7. Non-Functional Requirements (Quality Attributes)

Based on the user tasks and the tourism domain, four quality attributes are critical for STPS: Performance & Scalability, Security & Data Privacy, Usability & Accessibility, and Availability & Reliability. These attributes are aligned with the ISO/IEC 25010:2023 product quality model, the current successor to the ISO/IEC 9126 model used in Section 9.4. Performance & Scalability covers performance efficiency (time behaviour and capacity) and the scalability sub-characteristic of flexibility; Security & Data Privacy covers security (confidentiality, integrity and accountability); Usability & Accessibility covers interaction capability (learnability, operability and inclusivity); and Availability & Reliability covers reliability (availability, recoverability and faultlessness) (ISO/IEC 2023).

Each subsection links the attribute to the goals in Section 2.2 and the pain points in Section 3.1, then states measurable requirements; Section 9.4 summarises how each quality area is verified. Capitalised terms, such as Booking, refer to the domain entities and actors in Sections 3.2 and 6, and a Bookable Offering is any Accommodation, Vehicle or Food Establishment listing (the Services class in Section 6).

7.1 Performance & Scalability

The first goal in Section 2.2, one main digital platform for promoting Sarawak tourism, requires STPS to stay responsive under normal and peak loads, such as during the Rainforest World Music Festival. Slow searches, delayed Booking responses or outdated content would also undermine the third goal of trustworthy and current local information, and would leave the pain point of having no combined discovery and booking channel unresolved.

The system shall display the first page of search and browse results for Bookable Offerings, Destinations and Local Article Info within 3 seconds at the 95th percentile, with 10,000 active Bookable Offerings in the catalogue.

The system shall record a submitted Booking and show the Tourist its status within 5 seconds at the 95th percentile, excluding processing time at the external Payment Gateway Provider.

The system shall meet the response-time targets above with 500 concurrent active sessions, including sessions from Malaysia and from at least one overseas region. To allow for growth, it shall also meet them at double this load (1,000 concurrent sessions and 20,000 active Bookable Offerings) by adding server capacity, without changes to the software.

The system shall make saved changes to Bookable Offerings, Destinations, Local Article Info and availability slots visible to Tourists within 60 seconds and shall notify Local Service Providers of new Bookings for their Bookable Offerings within the same 60 seconds.

7.2 Security & Data Privacy

The fourth goal in Section 2.2, protecting personal information under the Malaysian PDPA, and the Ungoverned Personal Data Handling pain point require enforceable technical controls over personal data. These controls support Task 1 (Register & Manage Tourist Account) and requirements PR-12 and PR-13 in Section 8.1.

The system shall encrypt personal data in Tourist, Booking, Payment and Review records at rest and in transit, and shall not store full payment card details, which are handled only by the external Payment Gateway Provider.

The system shall restrict each role to its authorised functions and data. Local Service Providers shall access only the availability-slot and Booking-decision functions for their own Bookable Offerings, and only the Tourist data needed to decide on their pending Bookings.

The system shall log the user, time and action of every Management Staff or Local Service Provider access to Tourist personal data and of every Payment verification or refund and shall alert Management Staff within 15 minutes of detecting suspected unauthorised access.

The system shall record the purpose and legal basis for processing each Tourist's personal data, restrict processing to those purposes, and keep a timestamped consent record, captured through Task 1, wherever consent is required.

The system shall track each Tourist request to access, correct or delete personal data through to resolution. It shall alert the Data/System Admin to any access or correction request still open 14 days after receipt, so that the request can be completed within the 21 days allowed by the PDPA, subject to permitted exemptions and retention obligations.

7.3 Usability & Accessibility

The first goal in Section 2.2 and the Language and Currency Barriers pain point require that first-time Tourists, including international visitors and users with disabilities, can use the platform on common devices with minimal assistance.

The system shall enable at least 90% of first-time Tourists to complete a defined core task, such as finding and viewing an Accommodation, unaided within 3 minutes, measured in a usability test with at least 10 representative participants.

The system shall support all Tourist core tasks on smartphones, tablets and desktop or laptop computers at screen widths from 360 to 1,920 pixels without horizontal scrolling, with every control operable by touch, keyboard and screen reader.

The system shall present all Tourist core-task screens in English and Bahasa Malaysia, consistent with the language assumption in Section 2.3, and shall show prices in Malaysian Ringgit (MYR) with an optional indicative conversion into the Tourist's chosen currency.

7.4 Availability & Reliability

The first goal in Section 2.2 requires a dependable platform, especially for Booking and Payment records. These targets rely on the assumption in Section 2.3 that the required server, network and data repository are already available and, because overseas Tourists book at all hours, they apply 24 hours a day, seven days a week.

The system shall achieve at least 99.5% monthly availability for its core services (search and browse, Booking and Payment), excluding scheduled maintenance announced at least 48 hours in advance. This allows no more than about 3.6 hours of unplanned downtime in a 30-day month.

The system shall restore core services within 2 hours of any unplanned outage (recovery time objective) with no more than 15 minutes of data loss (recovery point objective).

The system shall never confirm Bookings beyond an availability slot's capacity, including when Bookings are submitted at the same time or when a Package Booking includes the slot's Bookable Offering.

8. Other Requirements

8.1 Product-Level Requirements

Product-level requirements describe the capabilities that the completed Sarawak Tourism Promotion System (STPS) shall provide to support Tourists, Local Service Providers and authorised Management Staff. Each requirement is stated using “shall” to make it clear, testable and implementation independent.

PR-01 — Tourism Information Management

The system shall allow authorised users to create, view, update, publish, unpublish and archive tourism information and listings according to their assigned responsibilities.

PR-02 — Tourism Information Retrieval

The system shall allow Tourists to search for and retrieve relevant tourism information, including Destinations, Accommodation, Vehicles, Food Establishments, Packages and Local Article Info.

PR-03 — Information Categorisation

The system shall categorise tourism information according to defined tourism categories and attributes so that Tourists can locate relevant information efficiently.

PR-04 — Tourism Information Validation

The system shall validate required fields, data formats and applicable content conditions before tourism information, listings or promotional content can be published.

PR-05 — Availability Management

The system shall maintain availability information for each reservable tourism service and shall update the available capacity when applicable Booking events occur, including reservation, confirmation, cancellation, decline or staff adjustment.

PR-06 — Booking Management

The system shall allow authorised users to create, view and manage Booking records for Accommodation, Transportation, Food & Dining and Travel Package reservations.

PR-07 — Booking Decision Support

The system shall allow an authorised Local Service Provider to review and approve or decline pending Bookings for its own Bookable Offerings, and shall escalate unresolved Booking decisions to authorised Management Staff when the defined decision period expires.

PR-08 — Promotional Content and Listing Lifecycle Management

The system shall allow authorised Management Staff to create, update, publish, schedule, unpublish and archive tourism listings and promotional content according to their assigned responsibilities.

PR-09 — Travel Package Management

The system shall allow Tourists to view available Travel Packages and, where supported, construct an itinerary by selecting eligible tourism components for a specified travel period.

PR-10 — Payment Support

The system shall initiate payment through the external Payment Gateway Provider, record the resulting payment status against the relevant Booking, and shall not store full payment card details.

PR-11 — Review and Feedback Management

The system shall allow eligible Tourists to submit reviews and general feedback, shall support screening and moderation of submitted reviews, and shall allow authorised Management Staff to view and manage feedback according to the review policy.

PR-12 — Tourist Account Management

The system shall allow Tourists to register, authenticate and manage their account information, and shall restrict access to account information according to the user's authorisation.

PR-13 — Role-Based Access

The system shall restrict functions and information according to the authorised role of the current user.

PR-14 — Error Handling

The system shall provide an appropriate and understandable response when an operation cannot be completed, required information is missing or invalid, a requested service is unavailable, a Booking exceeds available capacity, or no matching information can be found.

PR-15 — Booking Status Management

The system shall maintain the current status of each Booking throughout its applicable lifecycle, including states such as Pending, Pending-Payment, Confirmed, Declined, Cancelled and Completed.

8.2 Design-Level Requirements

Design-level requirements specify constraints on interaction and presentation while remaining independent of a particular implementation technology (Sommerville 2016).

DR-01 — Consistent User Interface

The system shall use a consistent layout, terminology, navigation structure and interaction style across the major user tasks.

DR-02 — Clear Information Presentation

The system shall present tourism information using a clear information hierarchy so that users can identify important information without unnecessarily examining unrelated information.

DR-03 — Consistent Domain Terminology

The system shall use consistent terminology for domain concepts throughout the system.

For example, the terms **Tourist, Accommodation, Transportation, Food Establishment, Travel Package, Listing and Booking** shall retain consistent meanings throughout the system.

DR-04 — Clear Feedback and Error Messages

The system shall provide understandable feedback after important operations, including successful reservations, unsuccessful payments, unavailable services, declined Bookings and invalid information submissions.

DR-05 — Confirmation of Critical Actions

The system shall allow users to review or confirm significant actions before they are committed, particularly actions involving reservations, payments, cancellation and provider Booking decisions.

DR-06 — Accessibility-Oriented Presentation

The system shall present readable text, distinguishable interface elements, sufficient interaction feedback and keyboard-accessible controls to support users with different abilities and levels of digital experience.

DR-07 — Privacy and Terms Information

The system shall provide users with access to relevant privacy, terms-of-use and other applicable policy information before or during activities involving personal information or significant transactions.

DR-08 — Role Separation

The system shall separate functions and information presented to Tourists, Local Service Providers and authorised Management Staff according to their responsibilities.

DR-09 — Solution Independence

The requirements specified in this document shall remain applicable across the alternative solution approaches considered for STPS.

DR-10 — Maintainable Information Structure

The system shall organise tourism information so that authorised users can update individual listings, categories, promotional items or availability records without requiring unrelated tourism information to be modified.

9. Validation of Requirements

9.1 Purpose of Validation

Requirements validation was conducted to find whether the requirements accurately represent the needs of the intended users and stakeholders of the proposed Sarawak Tourism System. Requirements validation is crucial because errors or omissions in a requirements document can lead to substantial rework later in the software development process (Sommerville 2016). The validation process therefore focused on the completeness of the identified user tasks, correctness and consistency of domain terminology, consistency between actors, tasks and domain entities, identification of ambiguous, duplicated or contradictory requirements, consistency between functional and non-functional requirements, and verifiability of the specified requirements.

9.2 Requirements Review

The SRS was systematically reviewed by the project team. The review compared the task descriptions, domain model, workflows, quality attributes, product-level requirements and design-level requirements to identify inconsistencies and omissions.

Validation Check	Review Question	Finding Identified	Corrective Action
Completeness	Are the major Tourist, provider and Management Staff activities represented?	The main tourism discovery, reservation, provider and management activities are represented across Tasks 1–10.	Retained the 10-task structure and checked each task against the requirements.
Task consistency	Does each task's purpose agree with its subtasks?	Task 3's purpose includes identifying transportation and providing passenger/journey information, while its original subtasks did not explicitly cover all of these activities.	Align the Task 3 subtasks and workflow with the complete transportation reservation flow.
Task separation	Are similar tasks clearly distinguished?	Task 9 and Task 10 originally overlapped because Task 10's postcondition described updating daily capacity,	Keep Task 9 focused on availability and Task 10 focused on Booking decisions.

		which belongs primarily to availability management.	
Requirement wording	Are requirements expressed consistently and unambiguously?	Several original product/design requirements used “allows”, “provide”, “categorise” and other inconsistent forms.	Standardize formal requirements using “shall” statements.
Domain consistency	Are all entities used by tasks represented in the domain model?	Package and Add-on appear in the task/entity consistency material but are not listed in the main Section 3.2 entity list.	Add or clarify these entities in Section 6 before final submission.
Workflow consistency	Does each workflow represent the corresponding task?	Workflow steps need to match the final task subtasks after Task 3 is revised.	Review each Mermaid workflow against the final task table before submission.
Traceability	Can goals and pain points be traced to tasks and requirements?	The previous traceability table contained outdated task numbers, including Tasks 11–12.	Replace the traceability table with the corrected mapping in Section 9.5.
Verifiability	Can requirements be objectively checked?	Quality requirements already contain measurable targets, while several product/design requirements were previously too general.	Rewrite formal requirements using observable “shall” statements and specify validation methods.
Solution independence	Do requirements remain applicable across possible solutions?	The task descriptions generally describe user activities rather than specific technologies.	Retain implementation-independent wording and verify each solution against DR-09.

9.3 Task and Domain Entity Consistency Check

A task–entity consistency check was conducted to determine whether the activities identified in the task descriptions are adequately represented by the domain entities. The check helps identify missing or unnecessary relationships and supports traceability between requirements and the system model. Traceability is important because it maintains relationships between requirements and other software-development artefacts and helps identify gaps or inconsistencies in the requirements specification (Gotel & Finkelstein 1994).

Task	Tourist	Accommodation	Vehicle	Food Establishment	Package*	Booking	Review	Payment	Local Article Info
1. Discover Local Tourism Information	R	R	R	R	R				R
2. Reserve Accommodation	R	R				C/R/U			
3. Reserve Transportation	R		R			C/R/U			
4. Reserve Food & Dining	R			R		C/R/U			
5. Book Services	R	R	R	R	C/R/U	C/R/U		C/R	
6. Submit Review & Feedback	R	R	R	R	R	R	C/R/U		
7. Manage Promotional Content & Listing		C/R/U/D	C/R/U/D	C/R/U/D	C/R/U/D				C/R/U/D
8. Manage Bookings Availability		R/U	R/U	R/U	R/U	R/U		R/U	

9. Manage Own Listing Availability		R/U	R/U	R/U					
10. Decide on Own-Listing Bookings	R	R/U	R/U	R/U		C/R/U			

9.4 Quality Attribute Validation

The quality requirements were reviewed against the selected software quality characteristics and their associated measurable acceptance criteria. The ISO 9126 quality model provides a framework for identifying software quality characteristics and evaluation criteria (Behkamal,

Quality Area	Requirement	Acceptance / Verification Method
Performance & Scalability	Search and browse results shall be displayed within 3 seconds at the 95th percentile under the specified catalogue and concurrent-session load.	Load testing using representative search and browse scenarios.
Performance & Scalability	Submitted Booking status shall be shown within 5 seconds at the 95th percentile, excluding external Payment Gateway processing.	Timed Booking transaction testing under representative concurrent load.
Security & Data Privacy	Personal data shall be protected in transit and at rest and full payment-card details shall not be stored by STPS.	Security inspection, configuration review, database inspection and penetration/security testing.
Security & Data Privacy	Users shall access only functions and data authorised for their role.	Role-based access-control testing using Tourist, Local Service Provider and Management Staff accounts.
Usability & Accessibility	At least 90% of representative first-time Tourists shall complete a defined core task within 3 minutes unaided.	Moderated usability test with at least 10 representative participants.
Usability & Accessibility	Core tasks shall operate from 360 to 1,920 pixels and support touch, keyboard and screen-reader interaction.	Multi-device interface inspection and accessibility testing.
Availability & Reliability	Core services shall achieve at least 99.5% monthly availability.	Uptime monitoring and monthly availability report.
Availability & Reliability	Core services shall recover within 2 hours with an RPO of 15 minutes.	Controlled recovery/failover test and recovery measurement.
Availability & Reliability	The system shall not confirm Bookings beyond available capacity, including simultaneous requests and Package Bookings.	Concurrent Booking and capacity-boundary testing.

9.5 Requirements Traceability Check

A requirements traceability check was conducted to confirm that the identified requirements could be related to the project goals, pain points and user tasks. Requirements traceability helps maintain relationships between requirements and other related development artefacts and can assist in identifying missing or poorly connected requirements (Gotel & Finkelstein 1994).

Goal	Related Pain Point	Related Tasks	Requirements	Quality Areas
Provide one main digital platform for promoting Sarawak tourism	Unverifiable / fragmented tourism information; No combined Discovery and Booking Channel	Tasks 1–6	PR-02, PR-03, PR-05, PR-06, PR-09, PR-10, PR-11	Performance & Scalability; Usability & Accessibility; Availability & Reliability
Give the tourism association control over content and data	Manual Content Updates; Ungoverned Personal Data Handling	Tasks 7–10	PR-01, PR-04, PR-07, PR-08, PR-13, DR-08, DR-10	Security & Data Privacy; Availability & Reliability
Provide trustworthy and current local information	Unverifiable / fragmented tourism information; Manual Content Updates	Tasks 1, 7 and 9	PR-02, PR-03, PR-04, PR-08, PR-14, DR-10	Performance & Scalability; Usability & Accessibility; Availability & Reliability
Protect personal information based on Malaysian PDPA requirements	Ungoverned Personal Data Handling	Tasks 6, 7, 8 and 10, with account controls supporting Tourist interactions	PR-10, PR-11, PR-12, PR-13, PR-14, DR-07, DR-08	Security & Data Privacy

10. Possible solutions

Centralized Web-Based Smart Tourism Platform

One possible solution for the Sarawak Tourism Promotion System (STPS) is with a **centralised web-based smart tourism platform**. The platform would provide tourists with a single digital access point for discovering Sarawak tourism information and services, while also providing authorized Management Staff and Local Service Providers with functions for managing listings, availability and bookings.

This solution is consistent with the current STPS requirements, which identify the need for a single platform covering tourism discovery, trip planning, booking, payment and review, while also supporting management of tourism content and service availability.

A centralized tourism information architecture is supported by Lopes et al. (2019), who describe a tourism information system based on a **structured and validated central source of information**, together with an information-management module and web services that can support different user applications. Their case study also demonstrates the use of modular architecture designed to support adaptation and future growth (Lopes et al. 2019).

Tourist interaction

Tourists would access STPS through a web interface using a modern browser. They would be able to browse or search for:

- Destinations
- Accommodation
- Vehicles/Transportation
- Food Establishments
- Local Information
- Bookings

After discovering suitable services, tourists could proceed to the relevant reservation task. The system would use the same central tourism information and availability information for accommodation, transportation, dining and packages.

Mobile-First Smart Tourism Application (Lim)

A second possible solution is a cross-platform mobile app for Android and iOS. Tourists would use it to discover Sarawak tourism information and make Bookings before and during their trip. Local Service Providers would use a provider mode to manage availability and respond to Booking requests. Management Staff would keep a browser-based back office for content management. All channels would share one central source of tourism, availability and Booking data.

This solution meets the same requirements as the web-based platform. The Section 4 tasks are solution-independent, and their Solution Agnosticism Checks already list mobile options, such as a mobile onboarding flow (Task 1) and a mobile itinerary planner (Task 6). It also supports the Task 2 variants for location-based suggestions and offline-saved information. Dorcic, Komsic and Markovic (2019) reviewed 126 articles published between 2012 and mid-2017 and found that mobile technologies are changing the behaviour of both tourists and tourism providers. This matches the STPS need to serve both groups through one channel.

Key differences from the web-based platform (Lim)

Tourists would download the app from the Google Play Store or Apple App Store and could browse as guests or sign in. As well as the web platform's search features, the app would:

Suggest nearby Destinations, Food Establishments and Local Information using the device location, where the Tourist permits it

Save local information, maps and Booking confirmations for offline use in low-coverage areas such as remote national parks and rural longhouses

Send push notifications to Tourists about Booking updates and to Local Service Providers about new Bookings, supporting the 60-second target in Section 7.1

Let Local Service Providers update availability and accept or decline Bookings from their phones (Tasks 9 and 10)

Pass payments to the external Payment Gateway Provider, so card details are never stored on the device or by STPS

There are trade-offs:

Tourists must install the app first, which may discourage short-stay visitors.

The Android and iOS versions must both be tested, and every update must pass app-store review.

Offline content can go out of date, so availability and prices would be re-checked online before a Booking is confirmed (Section 7.4).

Interactive Virtual Tourism

One possible solution for Task 1: Discover Local Tourism Information is an interactive virtual tourism guide within the STPS website or mobile application. Tourists can explore Sarawak destinations through interactive photos, videos, 360° views and virtual tours before deciding where to visit. This approach is supported by the Sarawak Government's Sarawak Digital Economy Blueprint 2030, which highlights the use of technologies such as 360° views, virtual reality (VR), augmented reality (AR) and 3D modelling to improve tourism experiences (Sarawak Government, 2021).

The system could provide virtual experiences for heritage sites, cultural attractions, nature areas and other points of interest. Tourists could select a location and explore its important features without physically visiting the place. This could help Tourists understand different destinations and decide which places they are interested in visiting.

Management Staff could add and update virtual tourism content, including images, videos and destination information. This would allow STPS to provide more engaging local tourism information while supporting the goal of helping Tourists discover Sarawak destinations through digital technology.

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